Fiscal Year:	FY 2004	**Task Last Updated:**	FY 06/09/2005
PI Name:	Fraden, Seth Ph.D.		
Project Title:	Nucleation of Colloidal and Protein Crystals in the Vicinity of a Metastable Critical Point		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02254	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	10/01/2001	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2386		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The key to testing the theoretical concept is the ability to vary the range of the interparticle attractive potential between the colloids. This will be done by adding non-absorbing polymer to the colloidal suspension. It is predicted that polymers produce an effective attractive potential between the colloids, known as the depletion potential. The range of the potential is predicted to be controlled by the size of the polymer. Our goal is to test the recent theoretical claim that crystal nucleation rates of colloids are greatly enhanced in the neighborhood of a metastable critical point. The theoretical explanation put forth for enhanced crystallization rates is that near the critical point, the crystal nuclei are coated by a dense colloidal fluid, which lowers the surface tension and thus the free energy barrier to crystallization. Although there have been numerous experimental studies of model colloids the mechanism of critical point assisted nucleation has not been experimentally confirmed. This new mechanism of crystallization is expected to be relevant to macromolecules, such as proteins, as well as to		

	colloids. If this theory is borne out by experiment then this will be of tremendous importance to the field of x-ray crystallography because the rate limiting step in protein structure determination is the crystallization process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We performed static light scattering experiments on mixtures of a protein (lysozyme) and polymer, poly(ethylene glycol) (PEG) in aqueous buffers. Employing a two component virial expansion for the chemical potentials, we show that pairwise interactions, i.e., truncating at the second virial level, cannot account for all our observations. From experiments we extracted mixed second and third virial coeffcients and compared to theory using equivalent hard sphere values as a test of the Asakura-Oosawa (AO) like depletion theory for the explaining the effect of adding non-absorbing polymer to protein solutions. Our measurements show that the effect of adding PEG to lysozyme solutions is incompatible with AO theory. Additionally, we measured metastable liquid-liquid phase transition temperatures (cloud points) of these mixtures. Using the virial coeffcient values obtained from light scattering experiments we then calculated the predicted behavior of the cloud point based on a virial expansion of the free energy. Comparing results from the cloud point and light scattering measurements reveals that a virial expansion can qualitatively describe the system at low densities if mixed third virial coeffcients are included, but fails as the protein density is increased.
Bibliography Type:	Description: (Last Updated: 03/08/2019)
Articles in Peer-reviewed Journals	Bloustine J, Berejnov V, Fraden S. "Measurements of protein-protein interactions by size exclusion chromatography." Biophys J. 2003 Oct;85(4):2619-23. https://doi.org/10.1016/S0006-3495(03)74684-0 ; PubMed PMID: 14507724 ; PubMed Central PMCID: PMC1303485 , Oct-2003